

Day 5

Troubleshooting Network Services Practical (1 hour): Troubleshoot DNS and DHCP issues in a simulated network. Task: Document solutions for common DNS and DHCP issues.

1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point `www.spotify.com` to a wrong IP address, then try to ping the domain and document the error message you receive.

ANS :

1. Open **Notepad** as Administrator.
2. Open the hosts file:
C:\Windows\System32\drivers\etc\hosts
3. Add the following incorrect entry:
1.1.1.1 www.spotify.com
4. Save the file.
5. Open **Command Prompt** and run:

```
C:\Users\User>ping www.spotify.com

Pinging www.spotify.com [1.1.1.1] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 1.1.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Reason:

The hosts file overrides DNS resolution. Since the domain was mapped to an incorrect IP address, the system tried to contact the wrong server.

2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

ANS :

Configuration in Cisco Packet Tracer:

DHCP Pool Settings:

Setting	Value
Network	192.168.1.0/24
Default Gateway	192.168.2.1 (Incorrect)
DNS Server	8.8.8.8
Start IP	192.168.1.10

Client receives:

IP Address : 192.168.1.10

Subnet Mask: 255.255.255.0

Gateway : 192.168.2.1

Verification

Run:

ipconfig

Output:

IPv4 Address : 192.168.1.10

Default Gateway . . . : 192.168.2.1

Problem

- Client receives an IP address successfully.
- Internet and other networks cannot be reached because the gateway belongs to another network.

Solution

Correct the DHCP configuration:

Default Gateway = 192.168.1.1

Renew the IP:

ipconfig /release

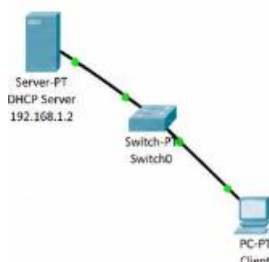
ipconfig /renew

Verify:

ipconfig

Now:

Default Gateway : 192.168.1.1



3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

ANS :

Scenario

Users can access websites using IP addresses but cannot access them using domain names.

Commands Used

ping 142.250.193.110

Result:

Reply from 142.250.193.110

Then:

nslookup www.google.com

Result:

DNS request timed out.

Server: Unknown

Request timed out.

Troubleshooting Steps

1. Check the configured DNS server.
2. Run : ipconfig /all
3. Verify DNS server address.
4. Test DNS server connectivity.
5. Restart the DNS service if required.
6. Flush DNS cache : ipconfig /flushdns
7. Try : nslookup www.google.com
8. Change DNS server to:
8.8.8.8 or 1.1.1.1

Solution

The DNS server address was incorrect. After changing it to a valid DNS server and flushing the DNS cache, domain name resolution worked correctly.

4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it. Hint : Check for network connectivity, DHCP scope exhaustion, and server availability

ANS :

Possible Causes

- Network cable disconnected
- Switch port disabled
- DHCP server is down
- DHCP scope exhausted
- DHCP service stopped
- VLAN configuration issue
- Firewall blocking DHCP traffic
- Incorrect DHCP configuration

Troubleshooting Steps

1. Check physical cable connection.
2. Verify NIC status.
3. Ensure the DHCP server is powered on.
4. Check available IP addresses in the DHCP pool.
5. Verify DHCP service is running.
6. Confirm VLAN configuration.
7. Release and renew the IP address:
 ipconfig /release
 ipconfig /renew
8. Test connectivity using:
 ping <DHCP Server IP>
9. Check Event Viewer for DHCP-related errors.
10. Restart the DHCP service if necessary.

Solution

After correcting the DHCP configuration and renewing the IP address, the client received a valid IP address such as:

IP Address : 192.168.1.15

Subnet Mask: 255.255.255.0

Gateway : 192.168.1.1

5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

ANS :

Service	Issue	Symptoms	Possible Cause	Recommended Solution
DNS	Incorrect DNS Server	Websites open by IP but not by name	Wrong DNS server configured	Configure a valid DNS server (e.g., 8.8.8.8)
DNS	Wrong Hosts File Entry	Incorrect website opens or ping fails	Invalid hosts file mapping	Remove or correct the hosts file entry
DNS	DNS Cache Corruption	Old or incorrect records	Outdated DNS cache	Run ipconfig
DHCP	APIPA Address (169.x.x.x)	No valid IP assigned	DHCP server unavailable	Check DHCP server, cable, and renew IP
DHCP	Wrong Default Gateway	Internet not accessible	Incorrect DHCP option	Configure the correct gateway in the DHCP pool
DHCP	Scope Exhaustion	New devices cannot get an IP	DHCP address pool is full	Increase the DHCP scope or remove unused leases